

Lab 3: Touch Sensor & Bar Graph

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Objective

The purpose of this lab was to operate a 10-LED bar graph using the capacitive touch sensing of the ESP32. The ESP32 used `touchRead()` to measure touch intensity after connecting a sensor wire to a touch input pin. The bar graph's LEDs glowed more when the touch was stronger. Additionally, the technology established a no-touch baseline for dependable detection via auto-calibration.

Materials

- 1× ESP32 microcontroller
- 1× Breadboard
- 10× LEDs
- 10× 220 Ω resistors
- Jumper wires (including **1 sensor wire** used as the touch pad)
- USB cable + Arduino IDE

Background and Theory

Capacitive touch sensing pins on the ESP32 are capable of detecting capacitance variations. The ESP32 can use `touchRead()` to measure a touch value when a sensor wire is connected to a touch-capable GPIO pin.

A “no-touch” wire has a baseline capacitance, but when a finger touches or comes close to the wire, the capacitance changes. In contrast to baseline, this typically results in a decrease in the `touchRead()` value. The capacitance changes more with increased surface contact (stronger touch, larger finger area, moist finger, many fingers), which causes the touch value to tend to decrease further.

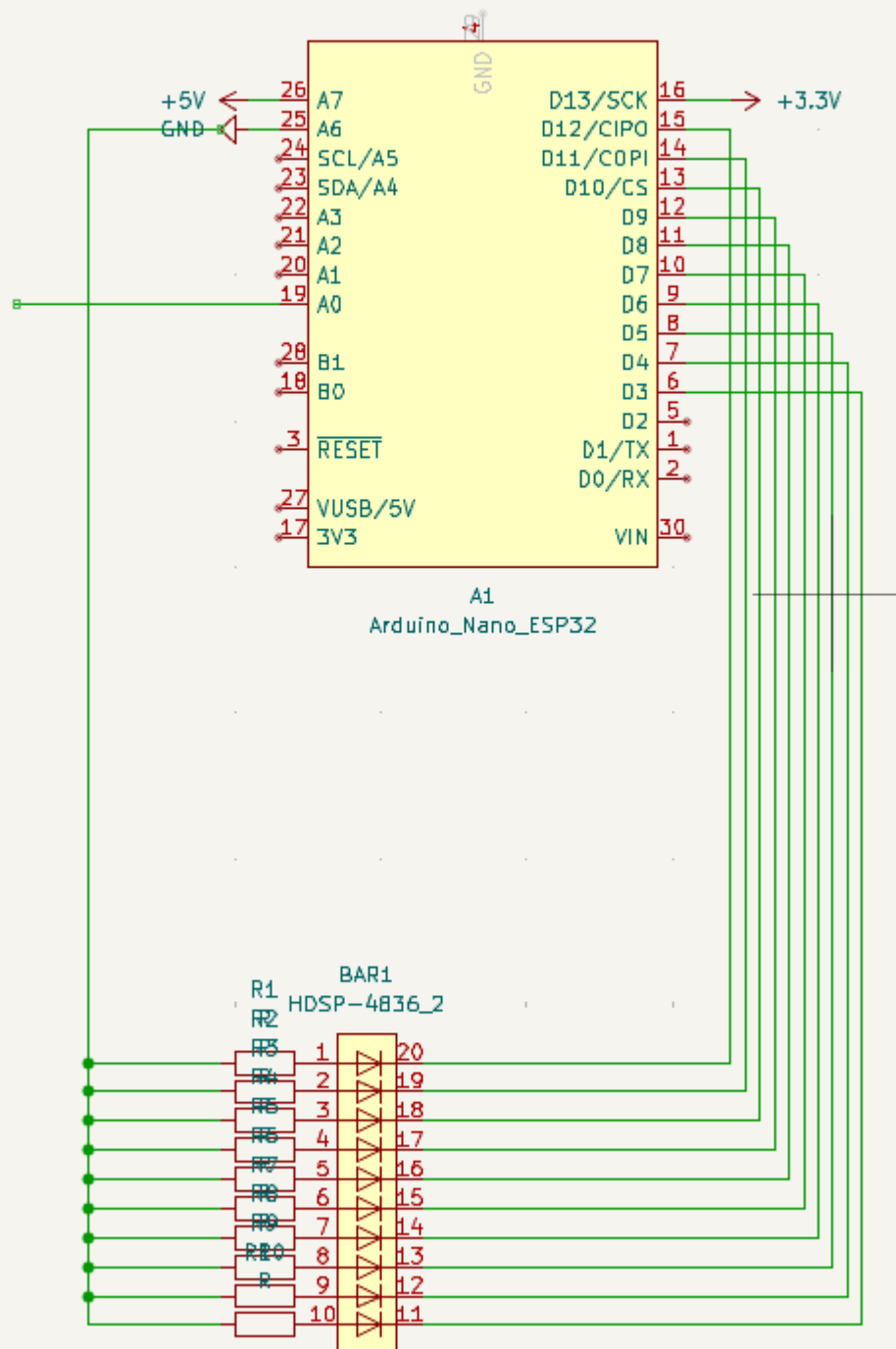
By sampling the no-touch value at launch, the system can automatically account for baseline readings that vary with the environment and sensor wires. A threshold can then be set slightly below baseline to detect touches consistently.

Wiring Configuration

- **Touch wire** → **GPIO 12** (touch input pin)
- **LED bar graph pins** → {} each through **220 Ω** to the LED anode
- **All LED cathodes** → **GND**
- **ESP32 GND** → **common ground rail**

Procedure

1. Connected the ESP32 to the breadboard and wired the 10 LEDs to the listed GPIO pins with 220 Ω resistors.
2. Connected one jumper wire as the touch sensor wire to the touch input pin (GPIO #).
3. Uploaded the program and opened Serial Monitor at **115200 baud**.
4. Allowed the ESP32 to auto-calibrate baseline while **not touching** the sensor wire.
5. Tested different touch types (tap, light touch, firm touch, longer touch).
6. Recorded Serial Monitor touch readings and the number of LEDs lit.



Code:

```
const int touchPin = 19;    // Touch sensor input pin

const int ledPins[10] = {15, 14, 13, 12, 11, 10, 9, 8, 7, 6};

// ----- Touch Calibration -----
int baseline = 0;
const int samples = 50;

// ----- Timing Control -----
unsigned long lastStepTime = 0;
const unsigned long stepInterval = 20; // ms per LED
int ledsOn = 0;

// Touch detection threshold
int touchThreshold = 0;

void setup() {
    Serial.begin(115200);

    for (int i = 0; i < 10; i++) {
        pinMode(ledPins[i], OUTPUT);
        digitalWrite(ledPins[i], LOW);
    }

    // Auto-calibration
    Serial.println("Calibrating... Do NOT touch the wire.");
    delay(2000);

    int total = 0;
    for (int i = 0; i < samples; i++) {
        total += touchRead(touchPin);
        delay(20);
    }
    baseline = total / samples;

    // Touch detected when value drops below this
    touchThreshold = baseline - 15;

    Serial.print("Baseline: ");
    Serial.println(baseline);
}

void loop() {
```

```

int touchValue = touchRead(touchPin);
unsigned long currentTime = millis();

bool touching = (touchValue < touchThreshold);

if (touching) {
    // Increase LEDs gradually while touching
    if (currentTime - lastStepTime >= stepInterval && ledsOn < 10) {
        digitalWrite(ledPins[ledsOn], HIGH);
        ledsOn++;
        lastStepTime = currentTime;
    }
} else {
    // Reset when not touching
    for (int i = 0; i < 10; i++) {
        digitalWrite(ledPins[i], LOW);
    }
    ledsOn = 0;
    lastStepTime = currentTime;
}

// Debug output
Serial.print("Touch: ");
Serial.print(touchValue);
Serial.print(" | LEDs: ");
Serial.println(ledsOn);

delay(20);
}

```

Results

Using the touch sensor, the LED bar graph was successfully controlled. The system reset the LEDs and kept the bar graph off when the sensor wire was left untouchable. LEDs started to light up when the wire was touched and the touch reading fell below the threshold. More LEDs illuminated as a result of stronger touches, which also caused greater variations in the touch reading.

Conclusion

Using a sensor wire and mapping the touch response to a 10-LED bar graph, this lab effectively illustrated the ESP32's capacitive touch sensing. Auto-calibration increased consistency under various settings and created a dependable no-touch baseline. Stronger touches led to more LEDs lighting up and larger variations in the touch reading, according to observations. All things considered, the lab reaffirmed embedded systems ideas such as multi-output control with a bar graph display, real-time input measurement, and sensor calibration.